

Paul Han

HARDWARE SYSTEMS ENGINEER | NEW PRODUCT INTRODUCTION (NPI)

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I've taken AI servers, nuclear weapons systems, and robotics data rigs from first schematic to mass production. I'll own design, bring-up, and validation while leading engineers, factories, and infra through NPI.

Education

B.S. Electrical Engineering & Political Science | Yale University | 2020 - AI and Analog Computing: Analog Crossbar Array CNN

Experience

HARDWARE FOUNDER | HAPTICA

2025-PRESENT

- **Tactile Glove:** Built a **1024-channel tactile-sensing glove** that records how skilled hands do physical work - training data for robot learning: piezoresistive ink formulation, plasma-etched flexible sensor skins, resistive-matrix crosstalk compensation, and FPGA error compensation on the analog read-out; prototyped magnetic and optical sensing alternatives
- **Egocentric Capture:** Paired the gloves with a **four-camera ZED/Jetson AGX Orin rig** - PTP hardware time-sync and supervised Linux capture services - producing synchronized tactile + visual streams ready for model training

HARDWARE ADVISOR | DEEPAI

2025-PRESENT

- Working advisor to the founder - evaluate a wide-ranging hardware portfolio, drive **build/kill decisions**, and own specifications, roadmaps, and engineering execution across Compute, FPGA, & Embedded programs.

HARDWARE SOFTWARE ENGINEER | LAWRENCE LIVERMORE NATIONAL LABORATORY

2023-2025

- Built the **electrical test and V&V platform** for the **W87-1 Joint Test Assembly** - the flight-test telemetry and sensing unit for a nuclear weapon - covering instrument control, test orchestration, and a custom test-head PCB with DC/DC, accelerometers, optical fiber transceivers, & FPGAs.
- Wrote a SpaceWire data-acquisition tool **100x faster** than the one it replaced (Python asyncio - ring buffer, CRC lookup table, vectorized NumPy), turning acquisition into a real-time design tool for the program; fed a Parquet/PostgreSQL/Tableau pipeline
- Developed a custom **inductance-to-digital gap sensor** and carried it through the program's first integrated delivery (EIT1)
- Held DOE Q clearance (2024-2025)

HARDWARE ENGINEER | META INFRA, OPEN COMPUTE PROJECT (OCP)

2020-2023

- Brought world-class **ML inference ASIC server platform** to market - owned test plans, component selection, electrical design refinement, and program management of Taiwan Original Design Manufacturer (ODM) through POC/EVT/DVT/PVT/MP
- Directed **OCP Debug Console** development as **Project Lead** - *Impact: Unlocked OOB debug for servers with bricked NIC during critical in-house NIC NPI, eliminating physical intervention during COVID*
- Released first complete specification of proprietary Meta debug interface as **Lead Writer** - *Impact: Improved debug functionality and cross-platform compatibility across all org products, created debug ecosystem*
- **Hands-on bring-up and lab debug** across x86 servers, ASIC power-on, power-rail sequencing, mixed-signal/SMPS circuits, USB and low-speed interfaces, and Linux; directed offshore test, debug, and design execution
- Returned after a 2019 electrical-test internship - built a first-of-kind **behavioral circuit simulator** in Python (state machines, graph traversal over OrCAD schematics) that replaced manual schematic verification

ELECTRICAL ENGINEERING INTERN | INTUITIVE SURGICAL

SUMMER 2018

- Developed **embedded PIC firmware** in C/C++ serializing time-of-flight and inductance-to-digital sensors over I2C and serial into a single data stream
- Designed **two PCAs** for next generation da Vinci Surgical System

Selected Hardware Projects

- **MiSTer FPGA Agent Eval:** Hardware-in-the-loop, verifiable-reward eval for frontier models - agents port MiSTer NES/SNES emulator cores to an Artix-7 M.2 FPGA built for closed-loop agent access: Vivado server, remote access, camera-verified gameplay, A/B multiboot with golden-image fallback - catching agents that report success without ever playing the game
- **Biometric Wearable:** Designed a PPG + IMU + temperature athletic wearable on the nRF52 platform - 6-layer layout with BGA, DFN, and 0201 packaging - plus **GaN capacitive wireless power**, **Tesla coils**, and a **ZVS induction heater**

Skills

Hardware: EDA - OrCAD, Altium, KiCad, atopile (code-defined PCB) · Power - DC/DC, SMPS, GaN, power sequencing · Signal integrity · Layout - 6-layer, flex · Platforms - OCP servers, x86, FPGA/CPLD · Buses - I2C/I3C, JTAG, PCIe, LVDS · Sensors - ToF, inductance-to-digital, IMU, optical, piezoresistive tactile

Software: C/C++, Python, Rust, Swift, TypeScript, SystemVerilog (Vivado) · Real-time DAQ - asyncio multithreading, ring buffers, PTP-synced multi-camera capture · Shipped apps - React Native, Swift macOS, Android · Postgres/TimescaleDB

AI / Agentic Engineering: Agent memory & retrieval (MCP servers in Rust) · Closed-loop agent↔EDA - Codex↔Vivado: synthesize, flash, verify on hardware · Multi-agent orchestration with review, test, and evidence gates · Agent-built test harnesses

NPI, Bring-up & Production: POC→EVT→DVT→PVT→MP phase ownership · Server/ASIC bring-up - power sequence, firmware, OOB/BMC, OS debug · Test strategy · DFM/DFT · Fixtures, SMD/BGA rework · Offshore factory and CM execution · Vendor qualification · Failure analysis · Production validation · Program management · BLE protocol capture and reverse engineering

Publications

Coauthor - IEEE APEC 2017 (capacitive-coupled wireless charging); IEEE EPE 2015 (induction-machine efficiency)